

BEATRICE, ON, Canada

WMO: 716940

Lat: 45.1361N

Lon: 79.3989W

Elev: 975

StdP: 14.19

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.3	-12.0	-24.2	1.5	-17.3	-17.8	2.1	-11.5	15.7	20.8	14.1	15.4	2.4	80	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	83.4	69.7	80.9	68.1	78.4	66.5	72.4	79.6	70.6	77.2	69.0	75.2	6.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	114.7	75.7	68.4	108.3	74.0	66.8	102.4	72.5	36.7	79.6	35.1	77.1	33.8	75.0	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.8	12.2	10.8	-29.9	88.0	5.6	2.2	-34.0	89.6	-37.3	90.9	-40.4	92.2	-44.5	93.8		
			-30.1	75.7	5.6	2.4	-34.1	77.5	-37.4	78.9	-40.6	80.2	-44.7	82.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.5	15.2	15.5	25.9	39.5	52.8	60.5	65.5	63.5	56.5	45.6	33.4	22.1	
	DBStd	20.25	13.90	12.26	12.34	8.86	8.47	6.18	5.05	5.26	7.62	8.07	9.59	11.57	
	HDD50	4776	1079	965	750	332	68	4	0	0	24	189	499	866	
	HDD65	8757	1544	1385	1211	766	391	162	55	90	272	603	947	1331	
	CDD50	1666	0	0	4	16	153	319	482	420	219	51	2	0	
	CDD65	174	0	0	0	0	12	27	72	45	18	0	0	0	
	CDH74	1858	0	0	3	10	174	303	767	444	153	4	0	0	
	CDH80	307	0	0	0	0	33	46	139	67	22	0	0	0	
Wind	WSAvg	4.5	4.7	5.0	5.1	5.4	4.7	4.1	4.0	3.8	3.8	4.5	4.4	4.6	
Precipitation	PrecAvg	40.6	4.1	2.8	2.7	2.6	3.2	2.9	3.6	2.9	3.9	4.1	4.2	3.9	
	PrecMax	47.3	6.4	4.5	5.5	4.8	5.3	4.0	6.4	4.5	5.6	7.0	6.7	5.8	
	PrecMin	33.3	2.4	0.9	0.8	0.8	1.1	1.3	1.0	0.9	0.8	0.9	2.3	1.4	
	PrecStd	3.4	1.1	1.0	1.3	1.0	1.2	0.8	1.5	1.0	1.2	1.7	1.3	1.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.9	46.2	65.9	75.2	83.8	84.2	87.1	86.0	83.3	74.0	63.0	49.5	
		MCWB	46.1	43.0	56.9	58.1	68.3	69.4	70.8	69.1	71.2	61.6	53.9	48.3	
	2%	DB	41.1	40.2	52.7	67.0	79.0	81.0	83.7	81.4	78.5	68.3	55.5	43.4	
		MCWB	39.8	38.1	44.6	51.5	65.3	67.4	70.5	68.7	68.3	60.5	49.8	41.9	
	5%	DB	37.0	35.6	47.1	60.7	74.4	77.5	81.0	78.8	74.5	63.9	51.7	39.9	
		MCWB	35.5	33.1	41.1	49.4	60.4	64.7	68.4	67.4	65.3	58.0	47.1	38.2	
	10%	DB	33.7	33.0	42.2	55.9	69.9	74.3	78.3	76.1	71.3	59.6	47.6	36.2	
		MCWB	32.5	31.1	37.3	46.5	57.3	62.7	66.5	65.8	63.7	55.0	44.8	34.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.7	44.0	58.0	59.8	70.3	73.3	75.4	73.8	72.5	66.7	56.0	48.9
			MCDB	46.9	44.3	66.7	67.3	81.0	81.5	83.3	80.6	80.1	68.9	60.3	49.9
2%		WB	40.0	38.3	47.0	55.7	67.1	70.1	72.6	71.6	70.2	61.9	51.1	42.1	
		MCDB	40.7	40.3	50.8	63.1	76.2	77.1	79.7	77.3	76.0	65.8	54.5	43.2	
5%		WB	36.0	33.7	41.6	51.3	63.6	67.6	70.6	69.7	67.9	58.9	47.8	38.3	
		MCDB	37.1	34.9	46.1	57.8	70.9	73.5	77.3	75.3	72.4	63.3	50.6	39.6	
10%		WB	32.6	31.3	37.9	47.7	60.5	65.6	69.0	67.8	65.2	55.6	44.8	34.9	
		MCDB	33.5	32.9	41.9	55.1	66.4	71.0	75.4	73.7	69.4	58.7	47.5	36.0	
Mean Daily Temperature Range	5% DB	MDBR	19.1	20.2	20.4	20.7	24.0	22.0	22.3	21.8	22.0	17.7	15.5	14.6	
		MCDBR	14.9	17.6	24.1	31.6	30.0	27.0	26.2	25.3	25.5	25.0	22.6	14.1	
		MCWBR	14.2	15.9	16.8	18.7	15.9	15.0	14.3	14.0	15.6	16.4	17.4	13.0	
	5% WB	MCDBR	15.0	17.0	20.8	26.5	24.1	21.4	21.3	19.6	20.9	20.7	19.6	13.4	
		MCWBR	14.8	16.5	16.0	18.0	14.5	13.6	12.7	12.3	14.0	15.3	16.6	13.0	
Clear-Sky Solar Irradiance	taub	0.283	0.288	0.309	0.349	0.372	0.399	0.402	0.394	0.368	0.340	0.306	0.279		
	taud	2.376	2.344	2.414	2.392	2.373	2.316	2.331	2.344	2.410	2.480	2.519	2.442		
	Ebn at Noon	264	285	294	289	284	276	273	272	270	263	256	255		
	Edh at Noon	23	29	32	36	37	40	39	37	32	26	21	20		
All-Sky Solar Radiation	RadAvg	420	700	1142	1425	1669	1821	1839	1587	1241	729	426	310		
	RadStd	51	89	119	128	163	144	129	90	120	78	54	37		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page